

NOTES

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Rationalizing the denominator

In general, you want to avoid radicals in the denominator. To rationalize the denominator, multiply both the numerator and the denominator by the conjugate of the denominator.

$$\begin{aligned}\frac{1}{1 + \sqrt{2}} \\&= \frac{1}{1 + \sqrt{2}} \cdot \frac{1 - \sqrt{2}}{1 - \sqrt{2}} \\&= \frac{1 - \sqrt{2}}{(1 + \sqrt{2})(1 - \sqrt{2})} \\&= \frac{1 - \sqrt{2}}{1^2 - \sqrt{2}^2} \\&= \frac{1 - \sqrt{2}}{1 - 2} \\&= \frac{1 - \sqrt{2}}{-1} \\&= -1 + \sqrt{2}\end{aligned}$$

Exercises

1.

$$\begin{aligned}\frac{1 + \sqrt{8}}{\sqrt{9} - \sqrt{2}} \\&= \frac{(1 + \sqrt{8})(\sqrt{9} + \sqrt{2})}{7} \\&= \frac{\sqrt{9} + 3\sqrt{8} + \sqrt{2} + \sqrt{16}}{7} \\&= \frac{3 + 6\sqrt{2} + \sqrt{2} + 4}{7} \\&= \frac{7 + 7\sqrt{2}}{7} \\&= 1 + \sqrt{2}\end{aligned}$$

Partial fraction decomposition

We have something of the form:

$$\frac{1}{(x - 1)(3x - 2)}$$

and we want to decompose it into partial fractions:

$$\frac{1}{(x - 1)(3x - 2)} = \frac{A}{x - 1} + \frac{B}{3x - 2}$$

Multiplying both sides by $(x - 1)(3x - 2)$, we get:

$$1 = A(3x - 2) + B(x - 1)$$

Setting $x = 1$, we get:

$$1 = A(3(1) - 2) + B(1 - 1) = A$$

Setting $x = \frac{2}{3}$, we get:

$$1 = A\left(3\left(\frac{2}{3}\right) - 2\right) + B\left(\frac{2}{3} - 1\right) = -\frac{B}{3}$$

Solving for B , we get:

$$B = -3$$

So the partial fraction decomposition is:

$$\frac{1}{(x-1)(3x-2)} = \frac{1}{x-1} - \frac{3}{3x-2}$$

Exercises

1. Decompose $\frac{44x + 48}{(x+6)(x-3)(x+2)}$

$$\frac{44x + 48}{(x+6)(x-3)(x+2)} = \frac{A}{x+6} + \frac{B}{x-3} + \frac{C}{x+2}$$

$$44x + 48 = A(x-3)(x+2) + B(x+6)(x+2) + C(x+6)(x-3)$$

Set $x = 3$:

$$44 \cdot 3 + 48 = B \cdot 9 \cdot 5$$

$$\frac{180}{9 \cdot 5} = B$$

$$B = 4$$

Set $x = -2$:

$$44 \cdot -2 + 48 = C \cdot 4 \cdot -5$$

$$-40 = C \cdot -20$$

$$C = 2$$

Set $x = -6$:

$$44 \cdot -6 + 48 = A \cdot -9 \cdot -4$$

$$A = -6$$

2. Given that

$$\frac{-2x-9}{x+3}x = \frac{1}{x+3} + \frac{B}{x}$$

find B

$$\begin{aligned}\frac{-2x-9}{x+3}x &= \frac{1}{x+3} + \frac{B}{x} \\ (-2x-9) &= x + B(x+3) \\ -3x-9 &= B(x+3) \\ \frac{-3(x+3)}{x+3} &= B \\ B &= -3\end{aligned}$$

3. Given that

$$\begin{aligned}\frac{-9x+36}{(x-3)(x-6)} &= -\frac{3}{x-3} + \frac{B}{x-6} \\ -9x+36 &= -3(x-6) + B(x-3) \\ -6x+18 &= B(x-3) \\ \frac{-6(x-3)}{x-3} &= B \\ B &= -6\end{aligned}$$

Functions

Horizontal asymptotes of rational functions

Rational functions are of the form:

$$f(x) = \frac{ax^n + bx^{n-1} + \dots}{cx^m + dx^{m-1} + \dots}$$

To get the horizontal asymptotes, you first take the **dominant term**. It's the leading term of the polynomial in the denominator, *without the coefficient*. Here x^m .

Then, you divide every term in both numerator and denominator, and evaluate at $x \rightarrow \infty$.

Radical functions

Domain

All the values where the radicand > 0 for even degrees, $(-\infty, \infty)$ for odd degrees.

Exercises

1. Domain of $d(x) = \sqrt[7]{-2x + \frac{1}{2}} + 1$?

$$-2x + \frac{1}{2} >$$

Roots of rational functions

First factor numerator and denominator, then cancel out common factors, then set the numerator to zero and solve.

Exercises

1. Solve $f(t) = 0$ where $f(t) = \frac{7t^2 + 14}{t^2 - 25}$.

$$\frac{7t^2 + 14}{t^2 - 25} = 0$$

$$\frac{7(t^2 + 2)}{(t + 5)(t - 5)} = 0$$

$$t^2 + 2 = 0$$

$$t^2 = -2$$

2. Solve $f(x) = 0$ where $f(x) = \frac{x^2 + 2x - 15}{x^3 + 5x^2 - 9x - 45}$.

$$\frac{x^2 + 2x - 15}{x^3 + 5x^2 - 9x - 45} = 0$$

$$\frac{x^2 + 5x - 3x - 15}{x^2(x + 5) - 9(x + 5)} = 0$$

$$\frac{(x + 5)(x - 3)}{(x + 5)(x^2 - 9)} = 0$$

$$\frac{(x + 5)(x - 3)}{(x + 5)(x - 3)(x + 3)} = 0$$

$$1 = 0$$

The equation has no solution.

3. Solve $f(t) = 0$ where $f(t) = \frac{t-3}{t^2+6t+5}$

$$\frac{t-3}{t^2+6t+5} = 0$$

$$\frac{t-3}{t^2+t+5t+5} = 0$$

$$\frac{t-3}{(t+5)(t+1)} = 0$$

$$t-3 = 0$$

$$t = 3$$

Statistics

Variance

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Covariance

Covariance is the measure of the joint variability of two random variables.

$$\text{Cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Graphing

Graphing cubic curves with one distinct real root

If we are given one root of a cubic equation, we can use synthetic division to factor it.

Example

Given that $(x - 3)$ is a factor of $f(x) = x^3 + x^2 - 4x - 24$, sketch the curve of $y = f(x)$.

First, synthetic division:

	x^3	x^2	x^1	x^0
3	1	1	-4	-24
		3	12	24
	1	4	8	0

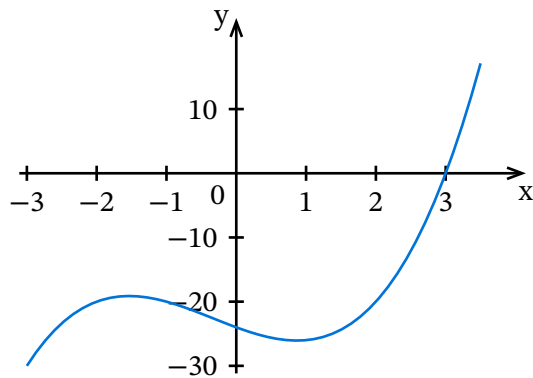
So we have $f(x) = (x - 3)(x^2 + 4x + 8)$

$\therefore x = 3$ is a root

The discriminant of $x^2 + 4x + 8$ is $4^2 - 4 \cdot 1 \cdot 8 = -16$, which is negative, so $x = 3$ is the only real root.

The y intercept is $f(0) = 0^3 + 0^2 - 4 \cdot 0 - 24 = -24$

Now we can draw the curve:



Logarithms

Domain

$(0, \infty)$

Range

$(-\infty, \infty)$

Trigonometry

Unit circle

Reference angle

The *reference angle* of an angle is the smallest angle between the terminal side of the angle and the x-axis. It is always a positive angle less than or equal to 90 degrees.

Pythagorean identity in the first quadrant

In the first quadrant, any point (x, y) forms a right triangle with the origin, where x and y are the lengths of the legs and $\sqrt{x^2 + y^2}$ is the length of the hypotenuse. Since in the unit circle, the length of the hypotenuse is 1, by definition, the Pythagorean identity states that $x^2 + y^2 = 1$.

Exercises

1. For a point in the first quadrant with $y = \frac{\sqrt{5}}{3}$, find its x coordinate.

$$\sqrt{\left(\frac{\sqrt{5}}{3}\right)^2 + x^2} = 1$$

$$\left(\frac{\sqrt{5}}{3}\right)^2 + x^2 = 1$$

$$x^2 = 1 - \left(\frac{\sqrt{5}}{3}\right)^2$$

$$x^2 = 1 - \frac{5}{9}$$

$$x^2 = \frac{4}{9}$$

$$x = \sqrt{\frac{4}{9}}$$

$$x = \frac{2}{3}$$

2. Point in the first quadrant. y coordinate is $\frac{\sqrt{2}}{3}$. What is the x coordinate?

$$\sqrt{\left(\frac{\sqrt{2}}{3}\right)^2 + x^2} = 1$$

$$\left(\frac{\sqrt{2}}{3}\right)^2 + x^2 = 1$$

$$x^2 = 1 - \left(\frac{\sqrt{2}}{3}\right)^2$$

$$x^2 = 1 - \frac{2}{9}$$

$$x^2 = \frac{7}{9}$$

$$x = \sqrt{\frac{7}{9}}$$

$$x = \frac{\sqrt{7}}{3}$$

Pythagorean trigonometric identity

In the unit circle, since for any central angle $x = \cos \theta$ and $y = \sin \theta$.

That means

$$x^2 + y^2 = 1$$

$$\cos^2(\theta) + \sin^2(\theta) = 1$$

This is the **Pythagorean trigonometric identity**. It works in all quadrants.

Exercises

1. Point P on the unit circle with $x = \frac{2}{5}$. Find $\sin \theta$.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta}$$

$$\sin \theta = \sqrt{1 - \frac{4}{25}}$$

$$\sin \theta = \sqrt{\frac{21}{25}}$$

$$\sin \theta = \frac{\sqrt{21}}{5}$$

2. y coordinate is $\frac{4}{9}$. Find $\cos \theta$.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\left(\frac{4}{9}\right)^2 + \cos^2 \theta = 1$$

$$\cos^2 \theta = 1 - \left(\frac{4}{9}\right)^2$$

$$\cos^2 \theta = 1 - \frac{16}{81}$$

$$\cos^2 \theta = \frac{65}{81}$$

$$\cos \theta = \sqrt{\frac{65}{81}}$$

$$\cos \theta = \frac{\sqrt{65}}{9}$$

3. x coordinate is $\frac{\sqrt{7}}{4}$. Find $\csc \theta$.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta}$$

$$\sin \theta = \sqrt{1 - \left(\frac{\sqrt{7}}{4}\right)^2}$$

$$\sin \theta = \sqrt{1 - \frac{7}{16}}$$

$$\sin \theta = \sqrt{\frac{9}{16}}$$

$$\sin \theta = \frac{3}{4}$$

$$\therefore$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\csc \theta = \frac{4}{3}$$

4. x coordinate is $\frac{3}{7}$. Find $\csc \theta$.

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \sqrt{1 - \frac{9}{49}}$$

$$\sin \theta = \sqrt{\frac{40}{49}}$$

$$\sin \theta = \frac{\sqrt{40}}{7}$$

$$\therefore$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\csc \theta = \frac{7}{\sqrt{40}}$$

$$\csc \theta = \frac{7}{2\sqrt{10}}$$

$$\csc \theta = \frac{7}{2\sqrt{10}}$$

$$\csc \theta = \frac{7\sqrt{10}}{20}$$

5. y coordinate is $\frac{1}{9}$. Find $\tan \theta$.

We have $\sin \theta = \frac{1}{9}$.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\cos \theta = \sqrt{1 - \sin^2 \theta}$$

$$\cos \theta = \sqrt{1 - \left(\frac{1}{9}\right)^2}$$

$$\cos \theta = \sqrt{1 - \frac{1}{81}}$$

$$\cos \theta = \sqrt{\frac{80}{81}}$$

$$\cos \theta = \frac{\sqrt{80}}{9}$$

$$\cos \theta = \frac{4\sqrt{5}}{9}$$

$\therefore$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\tan \theta = \frac{\frac{1}{9}}{\frac{4\sqrt{5}}{9}}$$

$$\tan \theta = \frac{1}{9} \cdot \frac{9}{4\sqrt{5}} \Rightarrow \tan \theta = \frac{\sqrt{5}}{20}$$

6. x coordinate is $\frac{\sqrt{11}}{6}$. Find $\cot(\theta)$

$$\text{We have } \cos \theta = \frac{\sqrt{11}}{6} \text{ and } \cot(\theta) = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta} = \frac{\frac{\sqrt{11}}{6}}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin^2 \theta = 1 - \left(\frac{\sqrt{11}}{6}\right)^2$$

$$\sin^2 \theta = 1 - \frac{11}{36}$$

$$\sin^2 \theta = \frac{25}{36}$$

$$\sin \theta = \sqrt{\frac{25}{36}}$$

$$\sin \theta = \frac{5}{6}$$

$\therefore$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\cot \theta = \frac{\frac{\sqrt{11}}{6}}{\frac{5}{6}}$$

$$\cot \theta = \frac{\sqrt{11}}{5}$$

Calculus

Limits

Limits of logarithmic functions

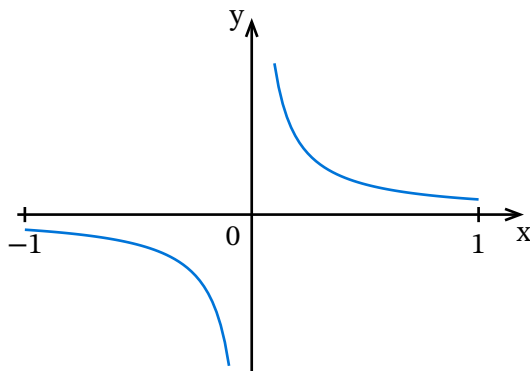
For any real n :

$$\lim_{x \rightarrow n} \log(x) = \log(x)$$

For infinity, remember the domain and apply transformations.

Limits of reciprocal functions

Remember, reciprocal functions have the form $f(x) = \frac{1}{x}$. The limit of a reciprocal function as x approaches infinity is always zero.



In general, for any non-constant polynomial $P(x)$, we have $\lim_{x \rightarrow \pm\infty} \frac{1}{P(x)} = 0$

But for $\lim(x \rightarrow 0)$, we have to consider the degree of the leading term of $P(x)$.

- Any function of the form $y = \frac{1}{x^n}$ where n is odd has the same shape as $y = \frac{1}{x}$. There is no limit at $x = 0$ because left and right limit are not the same.
- Any function of the form $y = \frac{1}{x^n}$ where n is even has the same shape as $y = \frac{1}{x^2}$. There is a limit, since the function will approach ∞ or $-\infty$.

Differentiation

Constant multiple rule

$$\frac{d}{dx}(kf(x)) = k \frac{d}{dx}(f(x))$$

Sum rule

$$\frac{d}{dx}(u(x) + v(x)) = \frac{du}{dx} + \frac{dv}{dx}$$

For example for $y = x^2 + x$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(x^2 + x) \\ &= \frac{d}{dx}(x^2) + \frac{d}{dx}(x) \\ &= 2x + 1 \end{aligned}$$

The power rule

Given a monomial $f(x) = x^n$, where n is a real number,

$$f'(x) = nx^{n-1}$$

Exercises

1. Calculate $\frac{dy}{dx}$ for $y = 2x - \sqrt{x}$

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(2x - \sqrt{x}) \\ &= \frac{d}{dx}2x - \frac{d}{dx}\sqrt{x} \\ &= 2 - \frac{d}{dx}x^{1/2} \\ &= 2 - \frac{1}{2} \cdot x^{-1/2} \\ &= 2 - \frac{1}{2\sqrt{x}} \end{aligned}$$

2. Find the derivative of $f(x) = \frac{1}{8}x^8$

$$\begin{aligned} f'(x) &= \frac{1}{8} \cdot \frac{d}{dx}x^8 \\ &= \frac{1}{8} \cdot 8x^7 \\ &= \frac{8x^7}{8} \\ &= x^7 \end{aligned}$$

3. Derivative of $f(x) = 8\sqrt{x}$

$$\begin{aligned}
 f'(x) &= \frac{d}{dx} 8\sqrt{x} \\
 &= \frac{d}{dx} 8x^{1/2} \\
 &= 8 \cdot \frac{d}{dx} \left(\frac{1}{2} \cdot x^{-1/2} \right) \\
 &= 8 \cdot \frac{1}{2} \cdot \frac{1}{2\sqrt{x}} \\
 &= \frac{4}{\sqrt{x}}
 \end{aligned}$$

Rate of change

Average rate of change on an interval $[a, a + h]$:

$$\frac{\delta y}{\delta x} = \frac{f(a + h) - f(a)}{h}$$

The *instantaneous rate of change* at a point $x = a$ is given by:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

Exercises

1. Given that $f(x) = x^2$, what is $f'(2)$ according to the definition?

$$\begin{aligned}
 f'(2) &= \lim_{h \rightarrow 0} \frac{(2 + h)^2 - 2^2}{h} \\
 &= \lim_{h \rightarrow 0} \frac{4 + 4h + h^2 - 4}{h} \\
 &= \lim_{h \rightarrow 0} \frac{4h + h^2}{h} \\
 &= \lim_{h \rightarrow 0} (4 + h) \\
 &= 4
 \end{aligned}$$

Integration

The antiderivative

The antiderivative is the opposite of the derivative.

Infinitely many antiderivatives map to the same function.

For example, since the derivative of x^2 is $2x$, we have x^2 as an antiderivative of $2x$. But $x^2 + 2$, $x^2 + 1$, etc. are also antiderivatives of $2x$.

In general, the antiderivatives of $2x$ follow the pattern $x^2 + C$ where C is a constant.

We can also formulate that using the $\int$ symbol for integration:

$$\int 2x dx = x^2 + C$$

Where C is called the **constant of integration**.

These integrals are **indefinite integrals**.

The function being integrated (here $2x$) is called the **integrand**.

The power rule for integration

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Exercises

1. Antiderivative of $\frac{1}{x^4}$?

$$\begin{aligned}\int \frac{1}{x^4} dx &= \int x^{-4} dx \\ &= \frac{x^{-4+1}}{-4+1} + C \\ &= \frac{x^{-3}}{-3} + C \\ &= -\frac{1}{3}x^3 + C\end{aligned}$$

2. Antiderivative of $x^{\frac{5}{8}}$

$$\begin{aligned}\int x^{\frac{5}{8}} dx &= \frac{x^{\frac{5}{8}+1}}{\frac{5}{8}+1} + C \\ &= \frac{x^{\frac{13}{8}}}{\frac{13}{8}} + C \\ &= \frac{8x^{\frac{13}{8}}}{13} + C\end{aligned}$$

3. Antiderivative of $\sqrt[5]{x^7}$

$$\begin{aligned}\int x^{\frac{7}{5}} dx &= \frac{x^{7/5+1}}{7/5+1} + C \\ &= \frac{x^{12/5}}{12/5} \\ &= \frac{5x^{12/5}}{12}\end{aligned}$$

4. Calculate $\int \left(\frac{z}{5}\right) dz$

$$\begin{aligned}\int \left(\frac{z}{5}\right) dz &= \frac{1}{5} \int z dz \\ &= \frac{1}{5} \cdot \left(\frac{z^2}{2} + C\right) \\ &= \frac{z^2}{10} + C\end{aligned}$$

The constant factor rule for indefinite integrals

To take the integral of a power function with a constant factor, like $5x^2$, we can take the constant factor out of the integral:

$$\int 5x^2 dx = 5 \cdot \int x^2 dx$$

This is the **constant factor rule**. In general:

$$\int kf(x)dx = k \int f(x)dx$$

Where k is a **constant**. It does not work if k is a variable.

Exercises

1. Calculate $\int \frac{1}{2y^5} dy$

$$\begin{aligned} \int \frac{1}{2y^5} dy &= \frac{1}{2} \int y^{-5} \\ &= \frac{1}{2} \cdot \frac{y^{-5+1}}{-5+1} + C \\ &= \frac{1}{2} \cdot \frac{y^{-4}}{-4} + C \\ &= \frac{1}{2} \cdot -\frac{1}{4y^4} + C \\ &= -\frac{1}{8y^4} + C \end{aligned}$$

2. Calculate $\int \frac{1}{3x^6} dx$

$$\begin{aligned} \int \frac{1}{3x^6} dx &= \frac{1}{3} \cdot \int x^{-6} \\ &= \frac{1}{3} \cdot \frac{x^{-5}}{-5} + C \\ &= \frac{1}{3} \cdot -\frac{1}{5x^5} + C \\ &= -\frac{1}{15x^5} + C \end{aligned}$$

3. Calculate $\int \frac{\pi}{2} dz$

$$\begin{aligned} \int \frac{\pi}{2} dz &= \frac{\pi}{2} \int z^0 dz \\ &= \frac{\pi}{2} \cdot z + C \\ &= \frac{\pi z}{2} + C \end{aligned}$$

The sum rule for integration

To take the integral of a sum, you can take the sum of the integrals of each term:

$$\int (f(x) \pm g(x))dx = \int f(x) dx \pm \int g(x) dx$$

The sum rule can be combined with the constant factor rule to integrate sums of constant multiples of functions:

$$\int (k_1 f(x) + k_2 g(x)) dx = k_1 \int f(x) dx + k_2 \int g(x) dx$$

Exercises

- Calculate $\int \sqrt{x}(2+x) dx$

$$\begin{aligned} \int \sqrt{x}(2+x) dx &= \int 2\sqrt{x} + x \cdot \sqrt{x} dx \\ &= \int 2\sqrt{x} dx + \int x \cdot \sqrt{x} dx \\ &= 2 \int \sqrt{x} dx + \int x^{3/2} dx \\ &= 2 \cdot \frac{x^{3/2}}{\frac{3}{2}} + \frac{x^{5/2}}{\frac{5}{2}} + C \\ &= \frac{4x^{3/2}}{3} + \frac{2x^{5/2}}{5} + C \\ &= \frac{4\sqrt{x^3}}{3} + \frac{2\sqrt{x^5}}{5} + C \end{aligned}$$

- Calculate $\int \frac{2+x^4}{x^2} dx$

$$\begin{aligned} \int \frac{2+x^4}{x^2} dx &= \int \frac{2}{x^2} dx + \int \frac{x^4}{x^2} dx \\ &= \int 2 \cdot x^{-2} dx + \int x^2 dx \\ &= 2 \cdot \int x^{-2} dx + \frac{x^3}{3} + C \\ &= 2 \cdot -x^{-1} + \frac{x^3}{3} + C \end{aligned}$$

- Calculate:

$$\begin{aligned} \int (3-x^2)^2 dx &= \int (9-6x^2+x^4) dx \\ &= 9 \int 1 dx - 6 \int x^2 dx + \int x^4 dx \\ &= 9x - 6 \frac{x^3}{3} + \frac{x^5}{5} + C \\ &= \frac{x^5}{5} - 2x^3 + 9x + C \end{aligned}$$

- Find the antiderivative of $z^3 + 2z - 1$

$$\begin{aligned}\int z^3 + 2z - 1 \, dz &= \int z^3 \, dz + \int 2z \, dz - \int 1 \, dz \\ &= \frac{z^4}{4} + 2\frac{z^2}{2} - z + C \\ &= \frac{z^4}{4} + z^2 - z + C\end{aligned}$$